

HWK 3

- 1) Calculate plate thickness of a quarter wave plate for light $\lambda = 585 \text{ nm}$. Refractive indices are $n_o = 1.534$ & $n_e = 1.543$.
Ordinary extraordinary

$$d_{\lambda/4} = \frac{(4m+1)\lambda_o}{4(n_o - n_e)} \rightarrow \frac{585 \cdot 10^{-9} \text{ m}}{4(1.534 - 1.543)} = \boxed{16.25 \mu\text{m}}$$

- 2) Calculate the thickness of a half-wave plate of $\lambda = 650 \text{ nm}$. Refractive indices are $n_o = 1.472$ & $n_e = 1.557$.

$$d_{\lambda/2} = \frac{(2m+1)\lambda_o}{2(n_o - n_e)} \rightarrow \frac{650 \cdot 10^{-9} \text{ m}}{2(1.472 - 1.557)} = \boxed{3.82 \mu\text{m}}$$

- 3) A beam of plane polarized light of $\lambda = 589 \text{ nm}$ normally on a thin piece of quartz. If $n_e = 1.5534$ & $n_o = 1.5443$, find thickness of quartz which will introduce a phase difference of 3π rad b/w e & o rays on emergence.

$$d_{\lambda/2} = \frac{((2(1)+1) \times \overset{589}{\cancel{589}} \cdot 10^{-9})}{2(1.5443 - 1.5534)} \rightarrow \boxed{97.887 \mu\text{m}}$$

- 4) A phase retardation plate of quartz has thickness $21.98 \mu\text{m}$. For what λ_o in the region $(400 \text{ nm} - 800 \text{ nm})$ will act as (a) quarter wave plate & (b) half-wave plate? Refractive indices are $n_o = 1.5534$ & $n_e = 1.5443$

Ⓐ $\lambda_o = d_{\lambda/4} \cdot 4(n_o - n_e) \rightarrow (21.98 \cdot 10^{-9}) \cdot 4(0.0091)$
 $\hookrightarrow \boxed{800 \text{ nm}}$

Ⓑ $\lambda_o = d_{\lambda/2} \cdot 2(n_o - n_e) \rightarrow (21.98 \cdot 10^{-9}) \cdot 2(0.0091)$
 $\hookrightarrow \boxed{400 \text{ nm}}$